

complex traversals visually.

Amazon Neptune supports querying through W3C's SPARQL and Neo4j openCypher to navigate through connected data sets with the support of Apache TinkerPop Gremlin. It also supports Open Graph API with Gremlin and SPARQL for fully-managed graphical data sets.

Recently, Amazon Neptune announced native integration and support for open source Graph Explorer for visually interpreting data sets through a web application interface based on ReactJS. It can be deployed as a Dockerized image in Amazon EC2 or Amazon ECS, and supports SPARQL and Apache TinkerPop Gremlin.

We can prepare data sets in Amazon Neptune through queries and use 'send to explorer' to connect to the Data Explorer source from Graph Explorer, in order to visually interpret the data sets and navigate the collection easily without querying data every time.

Graph Explorer reduces the work involved in querying and data set preparation, and is useful for non-technical users to query and interpret data visually through a drag-and-drop web UI interface. This can help in fraud detection, data grouping, and data mapping use cases with Big Data applications.

Google Dataplex: Data management solution on the GCP platform

Google Dataplex (Fig. 4) helps organisations centrally administer, discover, and monitor data across diverse data lakes and warehouses, and perform powerful data analytics.

The following are the key features of Dataplex.

Data discovery. It facilitates automated data discovery, classification, and enrichment of structured, semi-structured, and unstructured data in Google Cloud with powerful

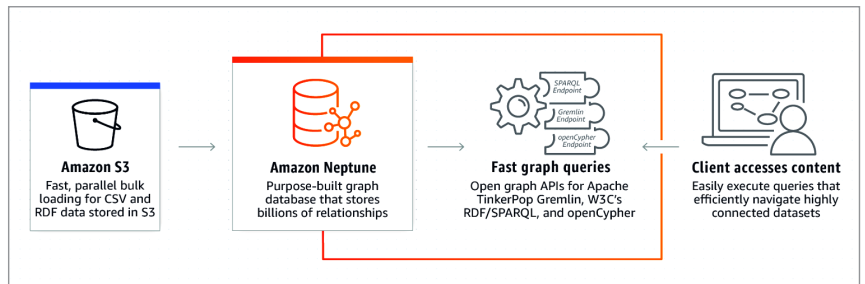


Fig. 3: Amazon Neptune

intelligence.

Data organisation. It supports organisation of data spanning multiple storage locations like data-plex lakes and zones.

Security. Supports powerful data enforcement policies, management, and monitoring of data for classification and authorisation.

Serverless data exploration. Supports interactive query generation with one-click access to SPARK-SQL queries and Jupyter Notebooks, enabling powerful collaboration between users.

Advantages of cloud data management

Cloud data management has a range of benefits.

Pay as you go. All cloud providers offer subscribers the benefit of paying on a per gigabyte per month basis. This helps organisations to cut excessive storage costs.

Scalability. Cloud providers provide strong and scalable solutions to subscribers. If an organisation runs short of storage at any time during the day or night, cloud providers automatically scale the storage and make sure that organisations never run out of capacity.

Flexible access. With cloud data management, subscribers can access their data anywhere and anytime.

Security. Cloud providers pro-

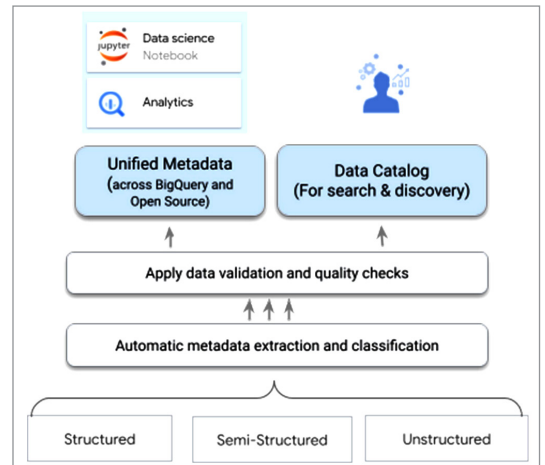


Fig. 4: Google Dataplex

vide an extra layer of security to keep the organisations' data safe and private. They honour all the security policies of the organisation.

Backup automation. Almost every cloud provider is rolling out services to automate the backup of crucial data at regular time intervals.

AI-driven enhanced intelligence. Modern cloud data management providers provide AI capabilities to ensure automatic discovery and cataloguing of data across various systems like ERP and CRM.

Cloud data management spans the end-to-end life cycle of data. Proper management minimises the risks and costs of security breaches. It also ensures that clients get accurate data when and where it is needed. **EFY**

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